Cambridge Assessment International Education

Cambridge International AS & A Level

BIOLOGY **9700/42**

Paper 4 A Level Structured Questions **May/June 2020**

MARK SCHEME

Maximum Mark: 100

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam. The answer column of the mark scheme shows the proposed basis on which Examiners would award marks for this exam. Where appropriate, this column also provides the most likely acceptable alternative responses expected from students. Examiners usually review the mark scheme after they have seen student responses and update the mark scheme if appropriate. In the June series, Examiners were unable to consider the acceptability of alternative responses, as there were no student responses to consider.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	'List rule' guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided • Any response marked *ignore* in the mark scheme should not count towards ***n*** • Incorrect responses should not be awarded credit but will still count towards ***n*** • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6	Calculation specific guidance Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form, (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	Guidance for chemical equations Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Mark scheme abbreviations:

;	separates marking points
/	alternative answers for the same marking point
R	reject
A	accept
I	ignore
AVP	any valid point
AW	alternative wording (where responses vary more than usual)
ecf	error carried forward
underline	actual word underlined must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument

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Question	Answer	Marks
1(a)	idea of shortage of, glutamate / an amino acid, in diet ;	1
1(b)	*any **two** from:* 1 uncontrolled mitosis / continuous cell cycle / cell cycle checkpoints not controlled ; 2 abnormal mass of cells formed ; 3 no programmed cell death / no apoptosis / cells immortal ; 4 AVP ; e.g. mutation of, tumour suppressor gene / (proto)oncogene	2
1(c)	osmoreceptors in hypothalamus ;	1
1(d)	*any **five** from:* 1 (ADH) secreted / released, into blood by posterior pituitary ; 2 binds to receptors on cell surface membranes ; 3 of collecting duct cells ; 4 enzyme cascade ; 5 vesicles with aquaporins move towards (cell surface) membrane ; 6 add aquaporins to (cell surface) membrane ; 7 water moves from collecting duct into blood, down a water potential gradient / by osmosis ; 8 water potential of blood rises ;	5

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(a)(i)</td><td>AAbb, Aabb, aaBB, aaBb ;</td><td>1</td></tr>
<tr><td>2(a)(ii)</td><td>
<table>
<tr><td></td><td>Alfresco</td><td>Di Nizza</td><td>or</td><td>Di Nizza</td><td></td></tr>
<tr><td>parent genotypes</td><td>aabb</td><td>Aabb</td><td></td><td>aaBb</td><td>;</td></tr>
<tr><td>gametes</td><td>ab</td><td>Ab ab</td><td></td><td>aB ab</td><td>;</td></tr>
<tr><td>offspring genotypes</td><td>aabb</td><td>Aabb</td><td></td><td>aaBb</td><td>;</td></tr>
<tr><td>offspring phenotypes</td><td>long</td><td>spherical</td><td></td><td>spherical</td><td>;</td></tr>
</table>
</td><td>4</td></tr>
<tr><td>2(a)(iii)</td><td>random / independent, assortment ;
metaphase 1 ;</td><td>2</td></tr>
<tr><td>2(b)(i)</td><td>any two from:
1 gourds might become invasive ;
2 more competition from gourds might reduce food production ;
3 might harm gourd genetic diversity ;
4 resistant gourds might have, unknown / adverse, effects on ecosystem ;</td><td>2</td></tr>
</table>

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Question	Answer	Marks
2(b)(ii)	*any **two** from:* 1 resistance gene makes hybrids more susceptible to *Erwinia* **;** 2 gene flow less of a problem than predicted (for this GMO) **;** 3 GM squashes are 'safe' to grow **;** 4 AVP **;** e.g. could still be a problem in areas lacking, beetle / *Erwinia*	**2**
2(b)(iii)	insecticides used to, kill beetle vector / prevent beetles spreading *Erwinia* **or** beetles may only feed on hybrids **;**	**1**

Question	Answer	Marks
3(a)	*any **four** from:* 1 *ref. to* geographical isolation**;** 2 no, gene flow / breeding, between populations (on the different islands) **;** 3 different, selection pressures / environmental conditions (on the different islands) **;** 4 different mutations occur (on the different islands) **;** 5 some mutations make individuals better adapted **;** 6 those individuals, survive / reproduce **;** 7 pass on advantageous alleles **;** 8 *ref. to* many generations **;** 9 reproductive isolation **;** 10 allopatric speciation **;**	**4**
3(b)	*any **two** from:* 1 bind to complementary base sequences in mtDNA **;** 2 (so) only amplify specific mtDNA sections / AW **;** 3 mtDNA section, differences / similarities, used to assess how closely related the subspecies **;**	**2**

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Question	Answer	Marks
3(c)	*any **one pair** from:* 1 large quantity in the cell ; 2 (so) easier to, extract / amplify, DNA for testing ; *or* 3 small genome size ; 4 (so) easier to locate specific section of DNA to test ; *or* 5 mtDNA is, a single copy of DNA / not paired alleles / haploid ; 6 (so) only mutation causes it to change ; *or* 7 inherited maternally ; 8 (so) all mtDNA sections are shared between all members (of maternal) family ; *or* 9 mutation rate is higher / no enzymes to repair mutations / faster molecular clock ; 10 (so)more choice of suitable sections of mtDNA to test / more accurate time estimate ;	**2**
3(d)	*any **two** from:* 1 habitat loss ; 2 more competition for, space / mates ; 3 reduction in prey ; 4 poaching / hunting / poisoning (by humans) ; 5 disease ;	**2**
3(e)	*any **two** from:* 1 morphological / physiological / behavioural / biochemical, similarity to mainland tigers / *Panthera tigris* ; 2 genetically similar to, mainland tigers / *Panthera tigris* ; 3 may be able to breed with, mainland tigers / *Panthera tigris*, to produce fertile offspring ;	**2**

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Question	Answer	Marks
4(a)	*any **three** from:* 1 repeat treatment causes, bigger / more rapid, immune response **;** 2 (so) target cells (expressing new protein) short-lived / AW **;** 3 decrease in (trans)gene expression on re-administration **;** 4 AVP **;** e.g. *ref. to* antibody binding / antibodies attract phagocytes	**3**
4(b)	*any **four** from:* *naked DNA* 1 has to be injected into target cell / lack of organ-specific delivery **;** 2 low efficiency of cellular uptake **;** 3 rapidly broken down **;** *viruses* 4 small packaging capacity / only small amount of DNA can be carried **;** 5 low probability of integration (into host genome) **;** 6 cause mutations in host DNA / (gene) insertion disrupts gene function / insertional mutagenesis **;** *liposomes* 7 low ability to, add DNA / genes, into target cells (genome) / low transduction efficiency **;**	**4**
4(c)(i)	1 red blood cells had 'self' antigens **;** 2 not recognised as foreign **;**	**2**
4(c)(ii)	(red blood cell) has no nucleus so no, DNA / transcription / translation, of targeting proteins **; ora**	**1**

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Question	Answer	Marks
5(a)	*any **three** from:* **transcription factor** / **promoter** 1 (C, H, O, N and) S / (C, H, O, N and) P ; 2 protein / DNA ; 3 amino acids / nucleotides ; 4 peptide bonds / phospho(di)ester bonds ; 5 globular / double helix ;	**3**
5(b)(i)	1 (because) sequences already known ; 2 (because) large, number of base pairs / quantity of data, to search ;	**2**
5(b)(ii)	chromosome **D** as it contains the most (possible) Mig1-binding promoter sites ;	**1**
5(b)(iii)	1 Mig 1 (also) regulates genes involved in other pathways ; 2 there could be several copies of (one or more of) these five genes ;	**2**
5(c)	*any **three** from:* 1 production of a repressor protein ; 2 repressor protein binds to operator ; 3 prevents RNA polymerase binding to promoter ; 4 *lac* / structural, genes not transcribed ; 5 repressor only binds when, not bound to lactose / lactose absent ; 6 regulatory gene / gene I, codes for repressor (protein) ;	**3**

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Question	Answer	Marks
6(a)(i)	homeostatis / negative feedback **;**	**1**
6(a)(ii)	sarcoplasmic reticulum **;**	**1**
6(a)(iii)	*any **four** from:* 1 (Ca^{2+}) binds to troponin **;** 2 troponin changes shape **;** 3 troponin / tropomyosin, moves **;** 4 binding sites on actin exposed **;** 5 myosin heads, attach / form cross bridges **;** 6 myosin head tilts and pulls actin **;**	**4**
6(a)(iv)	*any **two** from:* 1 activates protein kinase **;** 2 acts as second messenger **;** 3 AVP **;** e.g. stimulates signaling cascade activates other (named) enzyme	**2**

Question	Answer	Marks
6(b)	*any **three** from:* 1 decreased uptake of glucose / decreased permeability to glucose ; 2 decreased glycogenesis ; 3 decreased glucose, oxidation / respiration ; 4 change in, fatty acids / fat, metabolism ; 5 AVP ; e.g. *ref. to* effect on glucagon	**3**

Question	Answer	Marks
7(a)	volume of carbon dioxide produced ÷ volume of oxygen consumed ; per unit time ;	**2**
7(b)(i)	25 ; 18 ;	**2**
7(b)(ii)	0.72 ;	**1**
7(b)(iii)	lipid / fatty acid ;	**1**

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Question	Answer	Marks
7(c)(i)	*any **four** from:* 1 ETC / electron transport chain, functions as normal **;** 2 H^+ / protons, pumped into intermembrane space **;** 3 proton gradient established **;** 4 most, H^+ / protons, diffuse through, DNP **;** 5 into matrix **;** 6 less / fewer, H^+ / protons, pass through ATP synthase **;** 7 less ATP produced **;**	**4**
7(c)(ii)	*any **two** from:* 1 tiredness **;** 2 weight loss **;** 3 increased, heart rate / breathing rate	**2**

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>8(a)</td><td>
<table>
<tr><td>5</td><td>1</td><td>8</td><td>6</td><td>2</td><td>4</td><td>3</td><td>7</td></tr>
</table>
;;

all correct = 2 marks
four in correct consecutive sequence = 1 mark
</td><td>2</td></tr>
<tr><td>8(b)</td><td>
<table>
<tr><th></th><th>Fungi</th><th>Plantae</th><th></th></tr>
<tr><td>method of nutrition</td><td>heterotrophic</td><td>autotrophic</td><td>;</td></tr>
<tr><td>storage polysaccharide</td><td>glycogen</td><td>starch</td><td>;</td></tr>
<tr><td>main component of cell wall</td><td>chitin</td><td>cellulose</td><td>;</td></tr>
</table>
</td><td>3</td></tr>
<tr><td>8(c)</td><td>bacteria ;</td><td>1</td></tr>
<tr><td>8(d)</td><td>RNA or DNA ;

single-stranded or double-stranded ;</td><td>2</td></tr>
</table>

Question	Answer	Marks
9(a)	*any **nine** from:* 1 DNA cut with restriction enzymes ; 2 (DNA) fragments placed in wells in gel ; 3 at cathode ; 4 current / electric field, applied ; 5 fragments negatively charged ; 6 move towards anode ; 7 gel acts as a molecular sieve ; 8 smaller fragments move, faster / further, than larger ones ; 9 current switched off ; 10 *ref. to* Southern blotting / AW ; 11 *ref. to* staining / (gene / DNA) probes, for visualisation ; 12 alleles have different positions on gel ; 13 AVP ; e.g. desired, DNA / allele, selected / increased, by PCR	**9**

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Question	Answer	Marks
9(b)	*any **six** from:* *if present* 1 enables early treatment ; 2 lifestyle changes ; 3 elective / preventative, mastectomy ; 4 regular check ups ; 5 prevents unnecessary prolonged suffering if discovered and treated early ; 6 prevents early death ; 7 AVP ; e.g. social or family advantage of preventing parent's early death *if not present* 8 removes worry ; 9 *ref. to* planning a family ; 10 AVP ; e.g. prevention / early diagnosis, cheaper than later treatment	**6**

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Question	Answer	Marks
10(a)	*any **seven** from:* 1 pigments arranged in light-harvesting clusters ; 2 photosystems ; 3 accessory pigments surround primary pigment ; 4 accessory pigment example ; e.g. chlorophyll b / carotene 5 primary pigment is chlorophyll a ; 6 reaction centre ; 7 accessory pigments absorb light <u>energy</u> ; 8 *ref. to* different wavelengths to chlorophyll a ; 9 pass energy on to, chlorophyll a / primary pigment / reaction centre ; 10 pigments absorb light at different wavelengths / maximising light absorbed for photosynthesis ;	**7**

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Question	Answer	Marks
10(b)	*any **eight** from:* 1 carbon dioxide fixation / carbon dioxide combines with RuBP ; 2 using rubisco ; 3 6C unstable compound formed ; 4 2 x GP formed ; 5 GP reduced to TP ; 6 using ATP ; 7 and reduced NADP ; 8 from light-dependent stage ; 9 TP / GP, combines with, nitrate ions / ions containing nitrogen ; 10 ions enter via roots ; 11 ATP required ;	**8**

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